

1. RMP: Universal Reconstruction Mapping Protocol

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1.1. Abstract

Modern data distribution systems rely on transferring complete files, which causes inefficient bandwidth usage and unnecessary costs.

This paper introduces the **Reconstruction Mapping Protocol (RMP)**, a transfer paradigm that replaces raw byte transmission with a lightweight signed reconstruction map (.fmap).

Instead of sending full files, RMP sends instructions describing how to rebuild the target file through cached blocks, delta patches, procedural synthesis, and optional perceptual generation.

This enables substantial bandwidth reduction while preserving integrity and reconstruction guarantees.

1.2. Introduction

Traditional systems transfer complete files or static patches, ignoring structural redundancy and local availability.

RMP is based on a simple principle:

“Transfer structural knowledge instead of transferring all bytes.”

The protocol determines the most efficient reconstruction path using local resources and minimal downloaded data.

1.3. Core Components

1.3.1. Content Addressable Storage

Chunks are identified by cryptographic hashes. If a block already exists locally, it is reused.

1.3.2. Delta Encoding

If similar content exists locally, only binary differences are transferred.

1.3.3. Procedural Reconstruction

Deterministic generators may rebuild predictable structures.

1.3.4. Perceptual Mode

For media content, bounded approximation may be used when exact equality is unnecessary.

1.4. Security Model

Signed reconstruction maps Merkle verification of blocks **Zero-trust rebuild process** Sandboxed procedural modules

1.5. Cost Function

The protocol minimizes:

$$S = \alpha D + \beta C + \gamma(1 - P)$$

Where:

D = delta size C = compute cost P = cache probability

1.6. Evaluation Plan

Datasets:

Game assets Database dumps **Virtual machine images** Video archives Encrypted files

Metrics:

Transfer reduction Reconstruction time **CPU usage** Cache reuse ratio Accuracy

1.7. Conclusion

RMP proposes a new transfer model centered on structural orchestration rather than raw byte movement.

This approach can significantly reduce bandwidth usage in structured ecosystems while respecting information-theoretic limits.

1.8. References